

# KIPNGENO KEVIN

Software Engineer | Full-Stack Development | System Design

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## PROFESSIONAL SUMMARY

Software Engineer with experience delivering end-to-end software solutions, from architecture and design to development, deployment, and maintenance. Proficient in building scalable backend services, intuitive frontend applications, and cloud-based systems using modern engineering practices. Strong foundation in software architecture, system integration, API design, security, DevOps, and cloud deployment. Passionate about creating reliable, maintainable, and high-performance systems that align technical excellence with business objectives.

## TECHNICAL SKILLS

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|--------------------|-----------------------------------------------------------------------------------|
| Languages          | Java 21, JavaScript, TypeScript, Python, SQL, HTML5, CSS3                         |
| Backend            | Spring Boot, Spring Cloud, Node.js                                                |
| Frontend           | React, Next.js, Tailwind CSS                                                      |
| APIs & Integration | REST, SOAP, Webhooks, Kafka, HL7/FHIR (awareness), ERP integration                |
| Security           | OAuth2, JWT, RBAC, SAML, Spring Security, API Gateway patterns                    |
| Databases          | MySQL, PostgreSQL, Oracle, MongoDB                                                |
| DevOps & Cloud     | Docker, Kubernetes, GitHub Actions, Jenkins, AWS                                  |
| Testing            | JUnit, Mockito, Postman, TDD, SonarQube                                           |
| Compliance         | Kenya Data Protection Act, secure coding, RBAC enforcement, audit logging         |
| Methodologies      | Agile/Scrum, CI/CD, DevSecOps, Microservices, Modular Monolith, ADR-driven design |

## PROFESSIONAL EXPERIENCE

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**Software engineer** -Freelance | Jan 2025 – March 2025 | Nairobi, Kenya

#### Soinsync

- Developed and shipped web applications across real estate, fintech, and SaaS domains, building features from scratch while working within existing codebases and adapting to real project constraints.
- Diagnosed and resolved bugs across the full stack — tracing issues from frontend symptoms down to API logic and database queries — restoring system stability and preventing regressions through targeted fixes and test coverage.
- Improved application scalability by refactoring bottlenecks in API response times, optimising database queries, and restructuring code to handle increased load without compromising reliability.
- Maintained and extended existing software solutions — adding new modules, updating dependencies, and adapting systems to changing business requirements without breaking existing functionality.
- Restructured legacy and poorly organised codebases into clean, modular architectures, improving readability, reducing technical debt, and making systems easier to extend and maintain over time.
- Upgraded application security to modern standards — replacing weak authentication flows with JWT and refresh token rotation, enforcing role-based access control, hardening API endpoints, and ensuring financial workflows used idempotency to prevent duplicate transactions.
- Managed software upgrades including dependency updates, framework migrations, and environment configuration changes, ensuring systems remained stable, secure, and compatible with current tooling.

#### Software Engineering Intern

April 2024 – Aug 2024

#### Kobby Technologies Limited | Nairobi, Kenya

- Joined a professional engineering team to contribute to the design, development, and maintenance of cross-platform mobile applications integrated with ERP systems, gaining structured exposure to enterprise integration patterns under mentorship from senior developers.
- Implemented assigned RESTful API endpoints and integrated third-party services under code review, learning production coding standards, Git workflow conventions, and pull-request etiquette used by the engineering team.

- Shadowed and assisted in system analysis and requirements-gathering sessions, learning how business requirements are translated into technical specifications and architecture decisions.
- Wrote and executed manual and automated test scenarios to validate functionality, supporting end-to-end user acceptance testing ahead of client releases.
- Fixed scoped bugs from the team backlog and verified fixes in staging environments, building familiarity with real-world debugging and QA handoff processes.
- Collaborated daily with backend, mobile, and QA engineers in a fast-paced, deadline-driven environment, strengthening communication and cross-functional teamwork skills.

## Software Engineer Apprentice | Qwarqo

Aug 2023 – Jan 2024 | Nairobi, Kenya

- Contributed to backend design and development of a property discovery platform, participating in architecture discussions and implementing scalable solutions.
- Supported migration to a Single Page Application architecture, improving system responsiveness and user experience.
- Wrote and executed test scenarios to validate functionality and support end-to-end and UAT processes.

## KEY PROJECTS

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### MobMoney -Mobile Money Transfer Platform

Java, Spring Boot, TypeScript, MySQL

[github.com/kipngenokey/MobMoney](https://github.com/kipngenokey/MobMoney)

- Architected a production-style mobile money platform that guarantees financial consistency under concurrency and partial failure, using idempotency keys, the Saga pattern for distributed transactions, double-entry bookkeeping, and structured logging for operational transparency.
- Designed the transaction model to safely handle retries, timeouts, and rollback scenarios without double-spending, patterns directly transferable to claims and payment reconciliation in HMIS/EMR billing modules.
- Integrated layered security controls throughout the development lifecycle to protect financial data, mirroring the data-sensitivity discipline required in healthcare systems.

### Real-Time Fraud Detection System

Kafka, Spring Boot, REST, Machine Learning

- Designed a real-time fraud detection pipeline consuming live transaction events from Kafka, applying rule-based scoring alongside an ML model exposed via REST API for anomaly classification.
- Built a functional monitoring dashboard surfacing flagged transactions in real time, demonstrating rapid prototyping and event-driven system design within a time-constrained hackathon environment.

### Subscription Tracker API

Nodejs, Javascript, MySQL

- Built a backend API for managing recurring subscriptions end-to-end, covering plan creation, billing cycle scheduling, automated renewal reminders, and usage tracking across multiple subscription tiers.
- Designed a scheduled job system to handle renewal notifications and subscription state transitions (active, grace period, expired, cancelled), reducing manual intervention for lifecycle management.
- Modelled the data layer in MySQL with normalised entities for plans, subscriptions, and billing cycles, exposing clean RESTful endpoints with DTO-based contracts decoupled from internal persistence.
- Implemented authentication and role-based access control restricting plan and billing operations to authorised roles, alongside input validation and centralised exception handling for consistent API error responses.
- Validated business rules, proration on plan changes, duplicate-subscription prevention, renewal edge cases through unit and integration tests, an approach directly transferable to recurring billing and patient-plan management in HMIS/EMR systems.

## EDUCATION

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### B.Sc. Software Engineering

*Upper Second Class Honours*

Cooperative University of Kenya

Relevant Coursework: Data Structures & Algorithms, Database Systems, Distributed Systems, Operating Systems, Computer Networks

*Final Year Project: Smart Parking Management System using Computer Vision for Number Plate Detection*

## CERTIFICATIONS & TRAINING

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Introduction to Linux – Linux Foundation  
Java Programming Masterclass – Udemy